TITLE PAGE

PROBLEM STATEMENT ID	**SIH26035**
PROBLEM STATEMENT TITLE	Development of a Software Program / Application for Generation of Test Reports for Non-Automatic Weighing Instruments (NAWI) as per OIML Recommendation R 76
THEME	**Miscellaneous**
PS CATEGORY	**Software**
TEAM ID	**Team ID**
TEAM NAME	**Team Name**

TulaSetu

तुला + सेतु — the bridge from the test bench to the certificate

We do not fill the 62-page OIML R 76-2 report.
We **compute** it — the form is only a rendering.

$$E = I + \tfrac{1}{2}e - \Delta L - L \quad \cdot \quad Ec = E - E_0 \quad \cdot \quad \text{pass if } |Ec| \le mpe$$

Ministry of Consumer Affairs, Food & Public Distribution · Legal Metrology Act, 2009 s.22 · LM (General) Rules, 2011, Seventh Schedule Heading A · LM (Approval of Models) Rules, 2011 · OIML R 76-1:2006 & R 76-2:2007

Team Name
TEAM

TULASETU — THE REPORT IS COMPUTED

Smart India Hackathon
2026

PROPOSED SOLUTION (DESCRIBE YOUR IDEA / SOLUTION / PROTOTYPE)

Detailed explanation of the proposed solution

- A **deterministic calculation engine** — every cell of the report is computed, never typed.
- 17 R 76-2 tests reduce to **6 kernels** + one multi-interval-aware **mpe resolver**: 36 form pages, 34 rules, 363 §17 items.
- Officer's path: profile → test list → observations → checklist → attachments → **PDF + DOCX + JSON**.

±0.5 e	±1.0 e	±1.5 e	×2
0 – 500 e	500 – 2000 e	2000 – 10000 e	in service

Class III, initial verification — Seventh Schedule Table 20 = R 76-1 Table 6.

How it addresses the problem

The application half of model approval is already digital on **eMaap**. From "model received at the RRSL" onward it is a spreadsheet — which the PS itself calls **time-consuming, error-prone and lacking uniformity**. We digitise that half.



Without the certificate there is no State licence — Act s.22, Rule 20. Every shop scale, jeweller's balance and weighbridge in India passes through this gate.

- **Errors:** the band is resolved per load *and* per partial range — the trap a spreadsheet falls into.
- **Uniformity:** "– –" carries the clause that excuses it, so no two RRSLs can disagree.

Innovation and uniqueness of the solution

- **Family / EUT solver (§3.10.4)** — the minimum variant set to physically test, writing the certificate's own "shall also cover…" clause.
- **Annex F module compatibility** — indicator + load cell checked on n, Emax, vmin, temperature overlap, $\Sigma p_i^2 \le 1$.
- **Dual conformity profile** — Seventh Schedule *or* OIML (Rule 7(1)(a)), with `IV ↔ IIII` normalisation since India's own rules use both.
- **Bilingual, offline, role-aware** — Hindi + English on the printed report; runs on one lab PC, no internet.
- **No AI in the legal path,** by design — OIML holds AI is not yet court-proof for legal metrology.

↕ The partial range changes here	The permissible error moves from ± 1 to ± 2 g. Nothing changes on the display (R 76-1 §3.3).						
6	4500	4512	0	13	**12.5**	3	⊗ FAILED
↕ The partial range changes here	The permissible error moves from ± 2 to ± 3 g. Nothing changes on the display (R 76-1 §3.3).						
7	5000	5000	0	1	0.5	3	⊘ PASSED

The working prototype, mid-test — row 6 fails (|Ec| 12.5 g > mpe 3 g, in red) and the permissible error steps ±1 → ±2 → ±3 → ±10 g as the load crosses each partial weighing range, while nothing changes on the instrument's own display. This is the case a spreadsheet gets wrong.

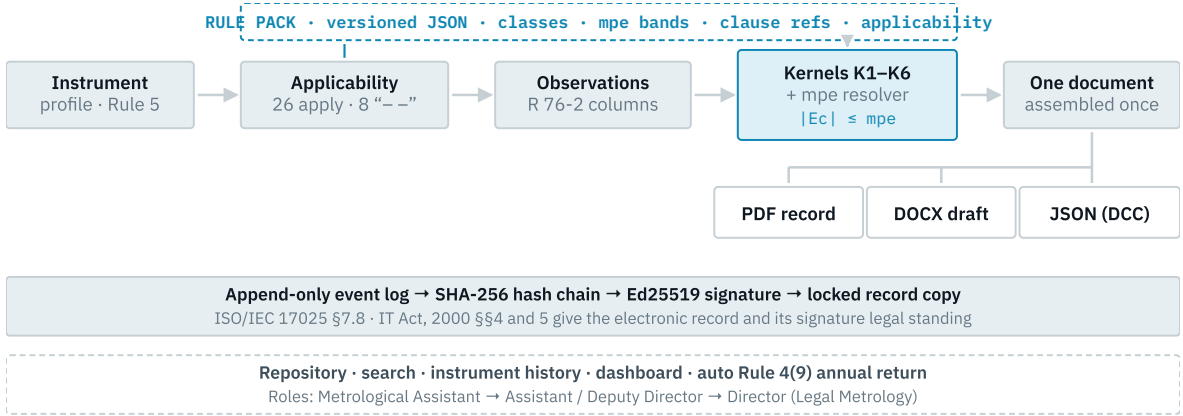
TECHNICAL APPROACH

Technologies to be used

`TypeScript` `React 19` `Vite` `Tailwind 4` `Node 24` `Fastify`
`SQLite → PostgreSQL` `JSON Schema` `Playwright → PDF` `docx` `Vitest`
`Ed25519 / SHA-256` `Tauri (offline desktop)`

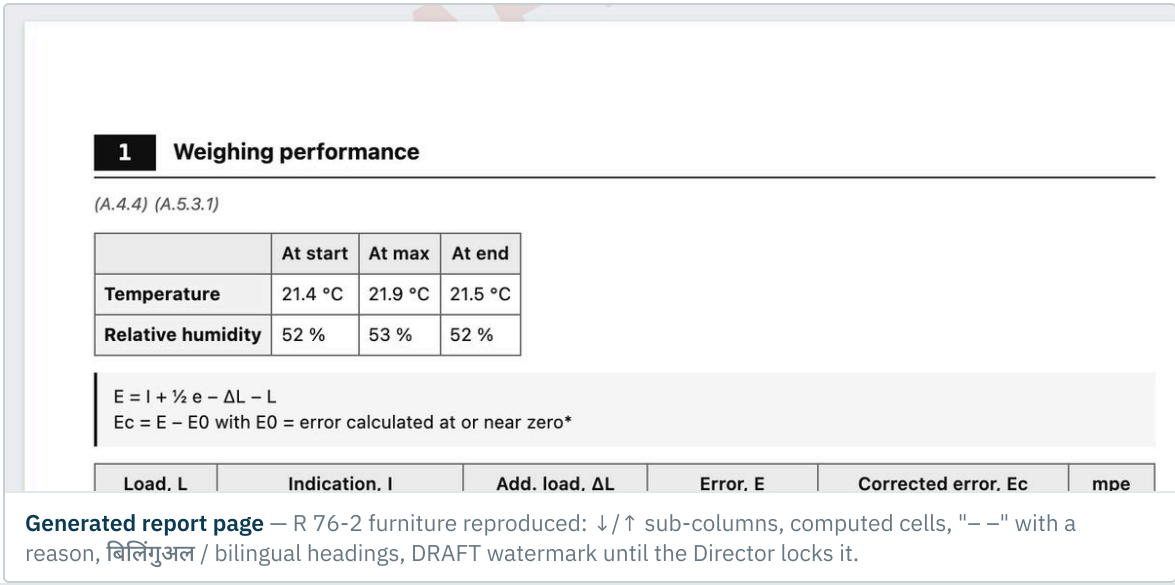
- **Engine = pure functions.** No I/O, no database, no randomness — which is what makes it fully unit-testable and reproducible.
- **Renderers never compute.** A renderer that needs a number not in the assembled document is a bug upstream — so PDF, DOCX and JSON cannot disagree.
- **Built for the Indian government stack.** NIC / MeghRaj hosting, CERT-In audit, GIGW 3.0 accessibility, DPDP-aware confidentiality tiers for Rule 16(2) drawings; eSign / DSC where the stub sign-in now sits.

Methodology and process for implementation





Applicability resolved from the profile — each excused form carries the clause that excuses it, e.g. "sensitivity is required only for non-self-indicating instruments (R 76-1:2006 §6.1)".



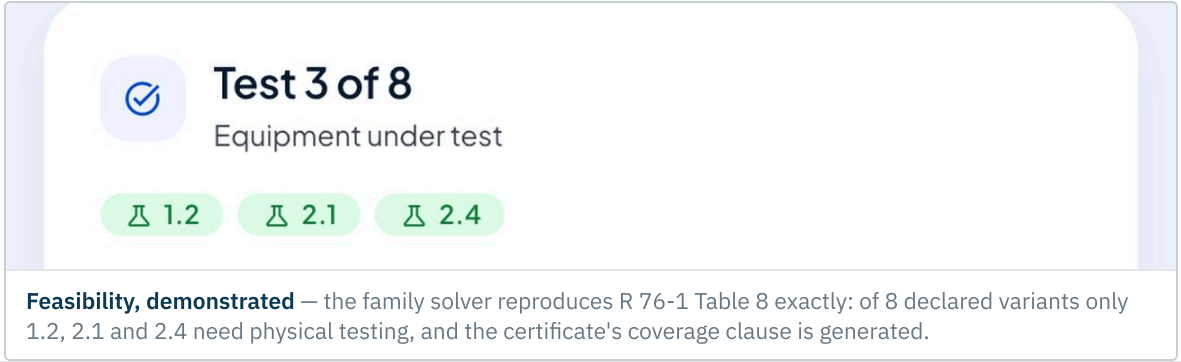
Generated report page — R 76-2 furniture reproduced: ↓/↑ sub-columns, computed cells, "– –" with a reason, बिलिंगुअल / bilingual headings, DRAFT watermark until the Director locks it.

FEASIBILITY AND VIABILITY

Analysis of the feasibility of the idea

332	36	363	3
unit & golden tests passing across engine, rule packs, renderers	R 76-2 form pages transcribed and rendered	§17 checklist items across 17.1–17.4	rule packs — OIML 2006, India Sch 7-A, 5-part draft

- **Already running, not a mock-up.** Every screenshot in this deck is the live application against its own engine.
- **Correctness is provable.** Golden tests come from the standards' *own* worked examples — R 76-1 §3.3's 2 / 5 / 15 kg multi-interval walk and Table 8's family — so a metrologist can check our output against the printed page.
- **Bounded scope, no procurement.** R 76-2 is a fixed, published format of 17 tests, and it runs on an ordinary lab PC, offline, in a browser.



Feasibility, demonstrated — the family solver reproduces R 76-1 Table 8 exactly: of 8 declared variants only 1.2, 2.1 and 2.4 need physical testing, and the certificate's coverage clause is generated.

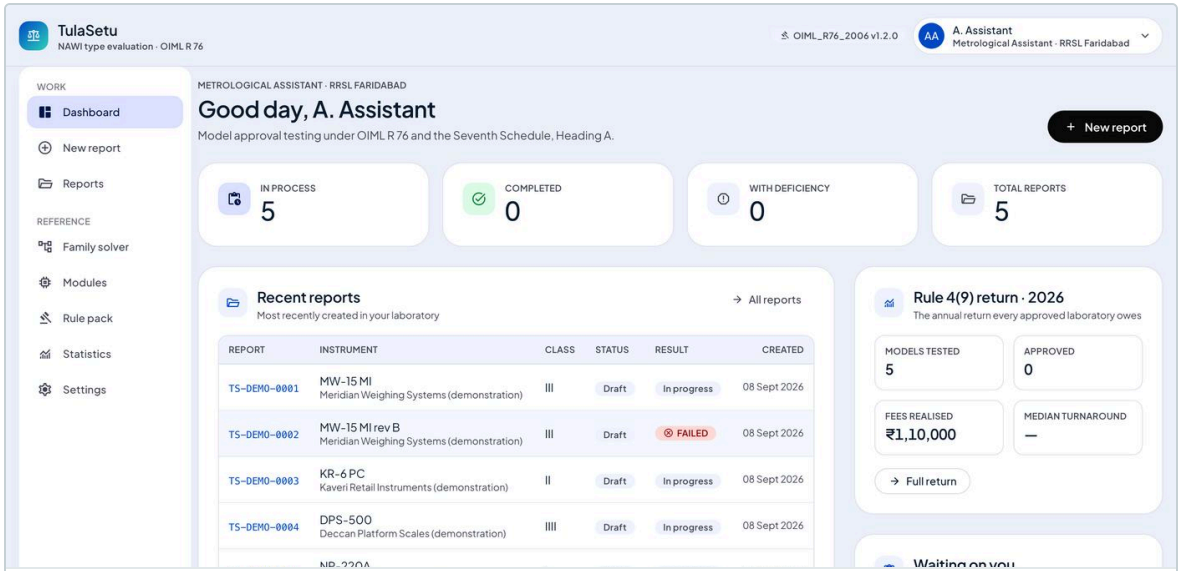
Potential challenges and risks · Strategies for overcoming them

RISK	STRATEGY
OIML R 76 is being rewritten from 2 parts into 5 — a hardcoded tool dies at the next edition	The standard is **data**. Three packs ship today and the engine resolves an identical permissible error under any of them; only numbering and citations differ. An edition change is a different file, not a different program.
India's Seventh Schedule must be followed, not just OIML — and its own text calls the ordinary class both `IV` and `IIII`	Dual conformity profile, which Rule 7(1)(a) expressly permits, plus class-symbol aliasing. Heading A was checked line by line against the raw legal text: Tables 17 and 20, the in-service ×2 factor and the multi-interval table are identical to OIML at source.
A legal record must survive challenge — an editable Word file cannot be the record	DOCX is a **DRAFT-watermarked review copy** naming the locked PDF's hash. The record copy is the hashed PDF in an append-only chain; IT Act, 2000 §§4–5 give it standing.
Judges and users expect "AI" in the answer	None in the legally relevant path, by design — OIML's own position is that AI is not yet court-proof for legal metrology. AI may only explain a deficiency in plain language, clearly outside the calculation.
Sign-in is a demonstration stub today	Stated openly in the product; the API refuses to start in production while it stands. Verification already takes a detached signature and a public key, so NIC single sign-on or a DSC-backed session drops in.
Adoption across the RRSLs, and by officers trained on paper	The screens read like the R 76-2 form officers already know. Rollout is a pilot at one RRSL, then the *rule pack* — not the code — is what each Director prescribes under Rule 7(2); training folds into the Indian Institute of Legal Metrology, Ranchi.

IMPACT AND BENEFITS

Potential impact on the target audience

- **RRSL officers** — the arithmetic and the re-checking disappear; the officer's judgement stays. Report preparation moves from days of spreadsheet work to the length of the bench test.
- **Assistant / Deputy Directors and the Director (LM)** — review a computed report with an audit trail, and the dashboard files the annual return that **Rule 4(9)** already demands: models tested, fees realised, time taken.
- **Indian manufacturers and MSMEs** — since 2023 India may issue OIML certificates, letting them export without paying for testing abroad. A correct R 76 report is what that certificate rests on; a rejection instead costs a fresh application and fresh fees under Rule 10.
- **Consumers** — every shop scale, jeweller's balance, ration-shop weigher and weighbridge reaches the market through this gate. A faster, uniform gate is a more trustworthy one.



Dashboard and repository — in-process / completed / deficiency counts, search and instrument history, and the Rule 4(9) annual return computed from the same log rather than re-typed.

Benefits of the solution (social, economic, environmental, etc.)

- **Economic — the second halving.** After model-approval services went online the end-to-end timeline fell from 6–9 months to 3–4 months (DoCA Annual Report 2022-23). Digitising the *application* produced that first halving; the report chain is what remains inside the 3–4 months.
- **Social — uniformity is fairness.** One engine and one versioned pack mean a manufacturer meets the same yardstick at every RRSL, and a citizen the same protection in every State.
- **Institutional — India's first structured NAWI test dataset.** Reports become machine-readable JSON instead of filing-cabinet PDFs, opening failure-pattern analysis by model family.
- **Regulatory fit.** Jan Vishwas moves legal metrology toward trust-based, facilitative enforcement; an append-only, signed record is exactly what a lighter-touch regime needs to stay auditable.
- **Environmental.** A 62-page report per model, printed and circulated for signature, becomes one signed electronic record.

ALIGNS WITH `SDG 1 · fair trade` `SDG 3 · clinical scales` `SDG 9 · quality infrastructure` `SDG 12 · honest quantity`

NOW · HACKATHON BUILD	NEXT · PILOT	THEN · SCALE
Engine, rule packs, wizard, three outputs, repository — running end to end.	One RRSL, eSign/DSC, eMaap and NSWS hand-off, C-DAC software-test intake for §17.4.	All RRSLs and GATC tenancies; in-service mode (mpe ×2) serves State re-verification.

RESEARCH AND REFERENCES

DETAILS / LINKS OF THE REFERENCE AND RESEARCH WORK

STANDARDS — THE MATHEMATICS AND THE FORM

- OIML R 76-1:2006 Metrological and technical requirements — Table 6 (mpe bands), §3.3 multi-interval worked example, §3.5.3.2 rounding, §3.10.4 + Table 8 families, Annex F modules, Annex G software · oiml.org/en/publications
- OIML R 76-2:2007 Test report format — the 17 tests and their printed forms
- OIML R 76 1.1 CD, Parts 1–5 (committee draft) — the 2 → 5 part restructuring our packs are tested against
- OIML D 31:2019 software for measuring instruments · B 18:2022 OIML-CS · ISO/IEC 17025:2017 §7.8

INDIAN LEGAL FRAMEWORK

- Legal Metrology Act, 2009 — s.22 model approval, s.24 verification; in force 1 April 2011
- LM (General) Rules, 2011 — Rule 13 → **Seventh Schedule, Heading A**; Rule 27 re-verification periods. Covers about 40 kinds of instrument, of which NAWIs are one.
- LM (Approval of Models) Rules, 2011 (+2019 amendment) — R.4(3) laboratory conditions, **R.4(9)** annual return, R.5 application contents, **R.7(1)(a)** Indian rules or OIML, R.8 report flow, R.10 resubmission fees, R.11(4) publication, R.19 fees, R.20 State licence gate
- LM (Government Approved Test Centre) Rules, 2013 — scope widened in October 2025 to 18 categories of instrument
- Jan Vishwas Acts, 2023 and 2026 — decriminalisation and improvement notices; Information Technology Act, 2000 §§4 and 5

HOW THE RESEARCH BECAME CODE

- Both standards were extracted to searchable text with page markers, so every threshold in a rule pack cites the page it came from and can be re-checked by grep.
- Golden tests were written from the standards' worked examples before the kernels — those examples are the acceptance criteria, not fixtures of our own.
- Heading A was compared line by line with OIML: Table 17, Table 20, the in-service ×2 factor and the multi-interval table are identical at the primary-source level.

PRIMARY INDIAN EVIDENCE WE GATHERED

- **16 real model-approval certificates** (2026) from the public register — the source of our field list and of the software-identification audit, where the checksum field is filled inconsistently from one certificate to the next
- DoCA Annual Report 2022-23 — the 6–9 → 3–4 month turnaround figure and the per-RRSL performance tables
- PIB, India's Evolving Metrology Ecosystem, 20 May 2026 — eMaap, One Nation One Time, India as the 13th country authorised to issue OIML certificates, GATC scope, Jan Vishwas
- eMaap portal — showing exactly where the digital application ends and manual work begins
- Indian Institute of Legal Metrology, Ranchi — the training route for officers

COMPARABLE PRACTICE AND PRIOR ART REVIEWED

- WELMEC 7.2 software guide · NMI (Australia) R 76-1:2015 · NCWM Publication 14 / VCAP · APEC NAWI verification handbook
- ISO/IEC 17025:2017 §7.8, reporting and amendment of results
- PTB digital calibration certificate (DCC) work — the model for our JSON output
- Published account of China's remote R 76 testing system — the basis for decoupling capture from report generation
- UNIDO, The role of metrology in the context of the SDGs

PROTOTYPE

- Live application: `add your deployed URL`
- Every threshold in the engine is recorded against its page of the standard in a verification log, so any number on screen can be traced to its printed source.